Product Requirements: Frontend Engineer

Owner: Fisayo Obadina | Project: SpendLens | Status: Draft | Date: 29 Oct 2026

1. Context

SpendLens is an API-first spending review product. Users import transaction records, review their spending, protect essential expenses, and test savings scenarios. Code does every calculation. AI only explains validated results. A demo app plays the role of a bank or fintech app embedding the API. Kora Checkout handles payment for paid features.

Team: Olatunji Oluwatobiloba (Team Lead, AI Engineer), Caleb Babatunde (ML Engineer), Fisayo Obadina (Frontend Engineer), Olamilekan Owolabi (Backend Engineer), Gift Oni (Business Analyst and presenter).

Free versus Pro (hypothesis, untested): Free gives one summary and a view-only scenario. Pro (30 days, test price set by an environment variable) adds saved scenarios, month comparison, goal progress, and later forecast and chat.

Timeline: v1 Oct 30 to Nov 6 (checkpoint Nov 7). v2 Nov 7 to Nov 13 (checkpoint Nov 14, lifeline). v3 Nov 14 to Nov 19 (checkpoint Nov 20). Feature freeze Nov 18. Rehearsal and backup video Nov 18 and 19.

Rules for everyone: synthetic data only. No banking passwords, no money movement, no automatic cancellations. AI never produces a number. Forecasts are ranges and the system abstains when data is thin. No invented users, revenue, partnerships, accuracy or compliance claims. Do not copy another company's brand or UI.

2. Your goal

Build a demo app that shows how a bank or fintech app would embed SpendLens, and lets a judge complete the whole workflow in a few minutes: import, review, protect essentials, test a savings cut, pay to unlock Pro.

Team decision to confirm: use a neutral fictional host app ("Partner Bank") instead of copying another company's UI and logo. Copying a real brand risks trademark and impersonation problems, and judges may read it as imitation. The point is to show the embed, and a neutral shell shows it just as well. Change this only if the team decides otherwise on purpose.

3. Users

Primary demo users: judges and early testers (students and young adults) using synthetic data. Secondary: a bank or fintech product person watching how the module would sit in their app.

4. Scope by version

Version	Screens and behavior
v1 (Oct 30 to Nov 6)	Partner shell with a "Money Insights" entry. Sample-data mode. CSV import wizard (choose file, preview, errors, confirm). Review screen with coverage summary, reconciliation candidates, category correction. Protect essentials. Scenario builder with calculation and supporting transactions. Plain-language explanation panel. Free versus Pro gate and Kora checkout. Delete my data.
v2 (Nov 7 to Nov 13)	Forecast view (range per category, with method and "not enough data" state). Recurring candidates list. Month-over-month comparison and goal progress (Pro).
v3 (Nov 14 to Nov 19)	Chat panel with streaming replies and evidence links. Voice input button (stretch, only if text chat is stable).

5. Requirements

- **R1.** The UI never calculates savings or totals. It displays numbers returned by the API.
- **R2.** Every finding links to the transactions behind it.
- **R3.** Uncertainty is stated in words ("Possible duplicate", "Not enough data"), not only color.
- **R4.** A first-time visitor can try the sample data without signing up.

- **R5.** Pro unlocks only after the backend reports the order as paid. A browser success callback is not proof of payment.
- **R6.** Every error has a clear message and a next step (bad CSV row, API asleep, payment failed).
- **R7.** Works on a phone-sized screen and with keyboard only.
- **R8.** Follow the Impeccable design guidance (github.com/pbakaus/impeccable) and avoid template-looking dashboards.

6. Acceptance criteria

- A new person completes import to scenario on sample data in under 3 minutes without help (test with someone outside the team).
- All numbers on screen match the API response exactly.
- Payment success, failure, cancel and pending states are all visible and tested.
- No keyboard traps. Visible focus. Labeled form fields. Contrast passes WCAG AA.
- Fixture mode lets the demo run if the backend is down.

7. Deliverables and dates

Date	Deliverable
Oct 30	Neutral shell, design tokens, fixture mode
Nov 1	Import wizard on fixtures
Nov 3	Review, reconciliation, category correction
Nov 4	Protections and scenario builder
Nov 5	Free versus Pro gate and Kora checkout (sandbox)
Nov 6	Integration with live API, accessibility pass
Nov 12	Forecast, recurring, comparison, goal progress
Nov 17	Chat panel, voice if stable
Nov 18	Feature freeze

8. Dependencies

API contract and OpenAPI file from Olamilekan by Oct 31. Fixture JSON for every endpoint (agree on it Oct 30). Kora sandbox public key from Olamilekan.

9. Out of scope

A native mobile app. A browser extension. Fake testimonials, fake trust badges, or any claim about users we do not have. Copying a real company's brand.

10. Risks

- Building screens before the API shape is settled. Use fixtures and agree on the contract first.
- Too many dashboard widgets. Show the one decision: what can I cut?
- Voice input depends on browser support. Treat it as optional.